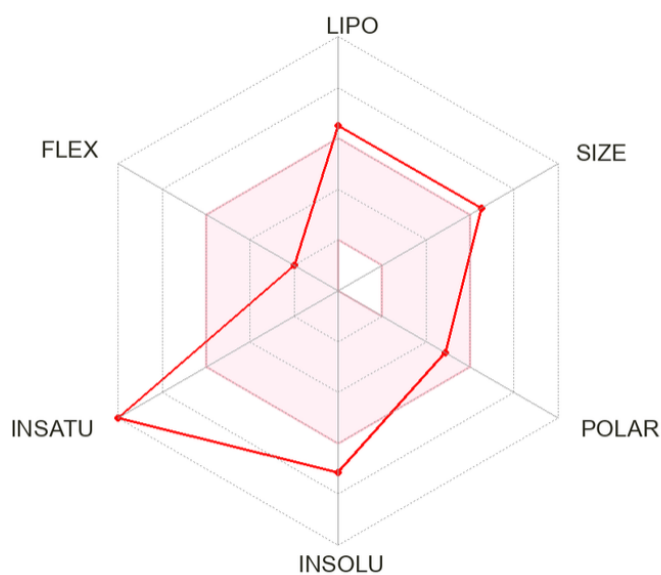
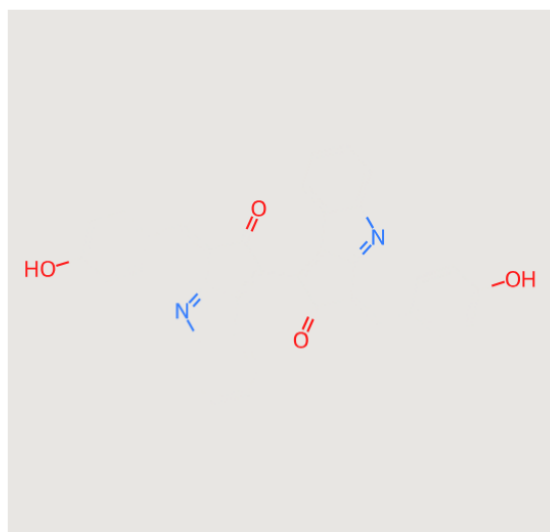


SwissADME Report

Compound: Scytonemin | **Formula:** C₃₆H₂₀N₂O₄

Source: SwissADME (swissadme.ch) — ADMET profiling, BRCA2 docking project



SMILES: Oc1ccc(cc1)/C=C\1/C(=O)C(=C2Nc3ccccc23)/C(=C\c2ccc(O)cc2)/C1=O

Physicochemical Properties

Molecular weight	544.55 g/mol
Formula	C ₃₆ H ₂₀ N ₂ O ₄
Num. heavy atoms	42
Num. arom. heavy atoms	24
Fraction Csp ³	0.00
Num. rotatable bonds	3
Num. H-bond acceptors	6
Num. H-bond donors	2
Molar Refractivity	170.46
TPSA	99.32 Å ²

Lipophilicity

Log Po/w (iLOGP)	Temporarily unavailable
Log Po/w (XLOGP3)	5.88
Log Po/w (WLOGP)	5.83
Log Po/w (MLOGP)	3.35
Log Po/w (SILICOS-IT)	8.15
Consensus Log Po/w	5.80

Water Solubility

Log S (ESOL)	-7.15
Solubility (ESOL)	3.90e-05 mg/ml ; 7.15e-08 mol/l
Class (ESOL)	Poorly soluble

Log S (Ali)	-7.74
Solubility (Ali)	9.94e-06 mg/ml ; 1.83e-08 mol/l
Class (Ali)	Poorly soluble
Log S (SILICOS-IT)	-11.17
Solubility (SILICOS-IT)	3.65e-09 mg/ml ; 6.70e-12 mol/l
Class (SILICOS-IT)	Insoluble

Pharmacokinetics	
GI absorption	Low
BBB permeant	No
P-gp substrate	No
CYP1A2 inhibitor	Temporarily unavailable
CYP2C19 inhibitor	Temporarily unavailable
CYP2C9 inhibitor	Temporarily unavailable
CYP2D6 inhibitor	Temporarily unavailable
CYP3A4 inhibitor	Temporarily unavailable
Log Kp (skin permeation)	-5.45 cm/s

Druglikeness	
Lipinski	Yes; 1 violation: MW>500
Ghose	No; 3 violations: MW>480, WLOGP>5.6, MR>130
Veber	Yes
Egan	Yes
Muegge	No; 2 violations: XLOGP3>5, #rings>7
Bioavailability Score	0.55

Medicinal Chemistry	
PAINS	0 alert
Brenk	1 alert: michael_acceptor_1
Leadlikeness	No; 2 violations: MW>350, XLOGP3>3.5
Synthetic accessibility	4.65